Uniformly Accelerated Motion

Kinematics of straight-line motion with constant acceleration · problem set 1 · student's copy

The rules of the game

For **every** problem, including the warm-up:

1. Choose the positive direction and set $x_0 = 0$ at the start.
2. Solve for what is asked.
3. Draw the three graphs against time: $a(t)$, $v(t)$, $x(t)$ — axes labelled, key values marked (initial/final speeds, times, distances).

Graphs need not be to exact scale, but the *shapes* must be right: with constant acceleration $a(t)$ is a horizontal line, $v(t)$ is a straight line with slope a, and $x(t)$ is a parabola (bending up if $a > 0$, down if $a < 0$). Use $g = 10\ \text{m/s}^2$.

The toolbox — the four equations of uniform acceleration:

$$\text{(i)} \quad v = u + at$$

$$\text{(ii)} \quad s = ut + \tfrac{1}{2}at^2$$

$$\text{(iii)} \quad v^2 = u^2 + 2as$$

$$\text{(iv)} \quad s = \frac{(u+v)}{2}\,t$$

u — initial velocity, v — final velocity, a — acceleration (constant), s — displacement, t — time. Choosing the right equation for each problem is part of the task.

Problems

0. **Warm-up (done in class — now add the graphs).** A train enters a station at speed v. The station has length L, and the train stops exactly at its far end, decelerating uniformly. Treating the train as a point, find the acceleration and the stopping time.

1. **Away from the lights.** A car starts from rest and accelerates uniformly at 2 m/s^2 for 6 s. Find its final speed and the distance covered.

2. **Between stations.** A train speeds up from 10 m/s to 25 m/s in 30 s at constant acceleration. Find the acceleration and the distance travelled.

3. **The cyclist.** A cyclist moving at 6 m/s accelerates uniformly at 2 m/s^2 along a 16 m stretch of road. Find his speed at the end of the stretch and the time it took.

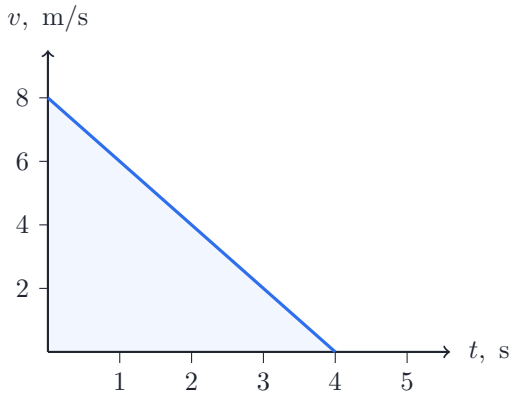
4. **Braking distance.** A car travels at 20 m/s. The driver brakes, and the car decelerates uniformly at 5 m/s^2. (a) How long and how far does it travel before stopping? (b) If the initial speed were doubled (40 m/s), how far would it travel? How many times longer is that?

5. **Off the bridge.** A stone is released from rest from a bridge and hits the water after 2 s. How high is the bridge, and how fast does the stone hit the water? (Ignore air resistance.)

6. **Straight up.** A ball is thrown straight up at 15 m/s. (a) How long does it take to reach the top? (b) What is the maximum height? (c) When does it return to the hand, and with what velocity? — On your graphs, pay special attention: what is the acceleration *at the very top*?

7. **The whole trip (graphs first!).** A bus leaves a stop: it accelerates uniformly from rest to 12 m/s in 4 s, drives at a steady 12 m/s for 6 s, then brakes uniformly to a stop in 3 s. Draw all three graphs *first*; then find the acceleration and the distance for each phase, the total distance, and the average speed of the trip.

8. **Read the graph.** The v–t graph of a tram is shown. From the graph, find the initial speed, the acceleration (slope), and the distance travelled (area under the graph). Then sketch the matching $a(t)$ and $x(t)$.



9. **Challenge: where equation (iv) comes from.** Using a v–t graph for uniform acceleration, show that the displacement equals the area of a trapezium: $s = \frac{1}{2}(u + v)\,t$. Then use this to re-derive your answer to Problem 0 in one line.